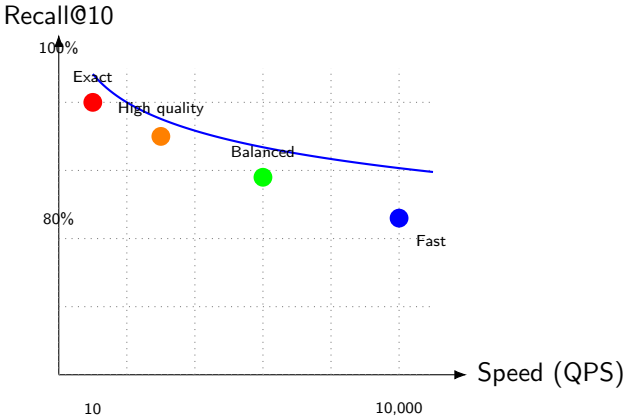
Full TikZ Figure Gallery

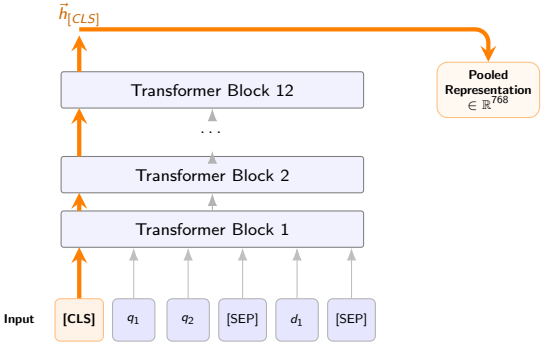
TikZ Migration System

March 9, 2026

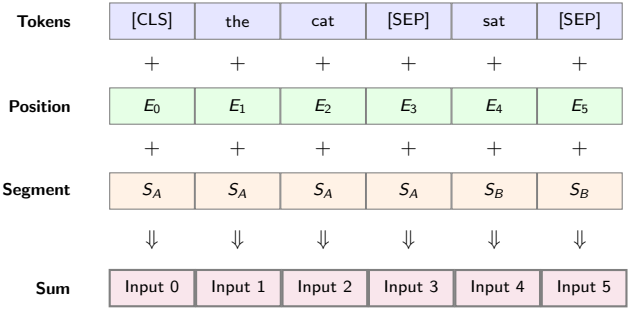
accuracy_speed_tradeoff.tex



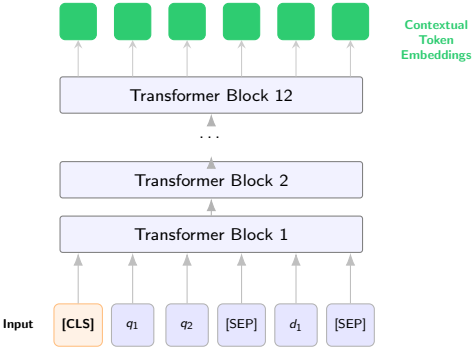
bert_cls_aggregation.tex



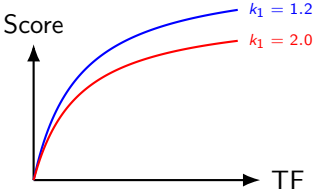
bert_input_process.tex



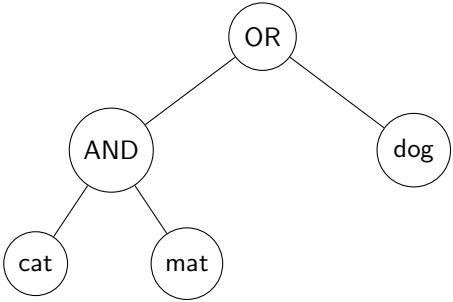
bert_token_representations.tex



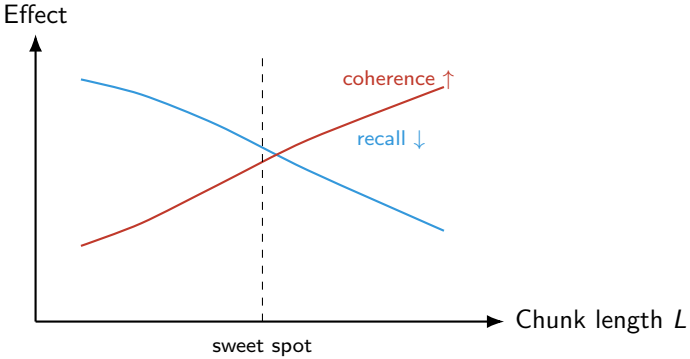
bm25_k1_comparison.tex



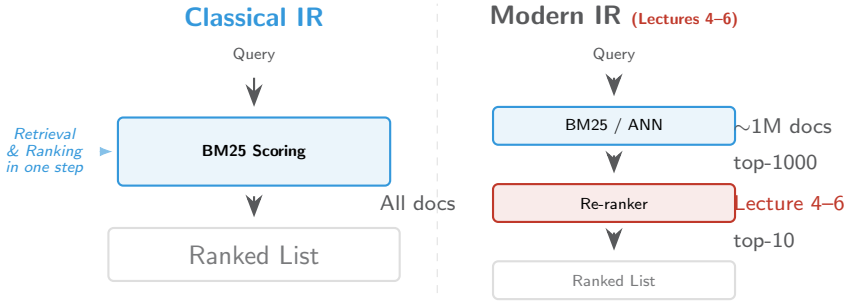
boolean_query_tree.tex



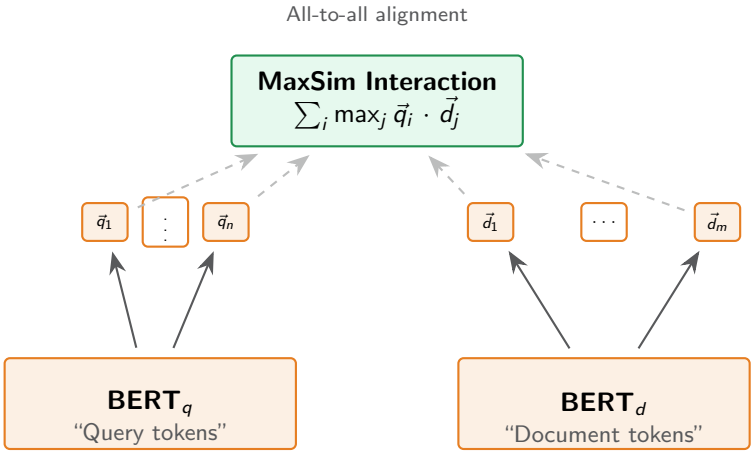
chunking_tradeoff.tex



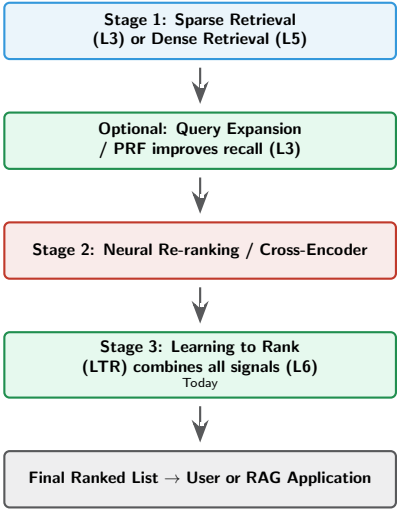
classical_modern_pipeline.tex



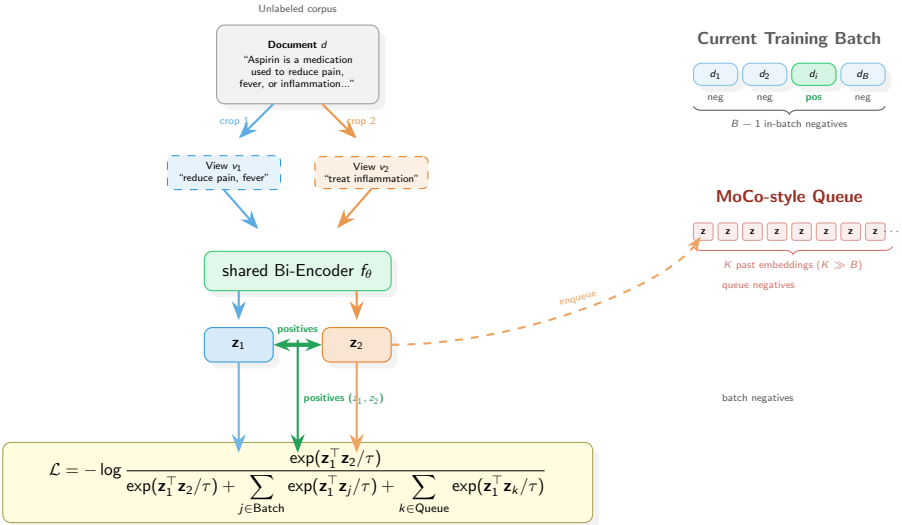
colbert_late_interaction.tex



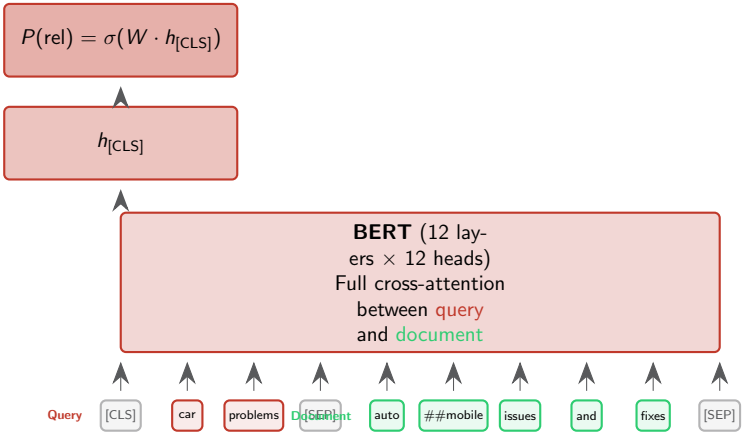
complete_retrieval_stack.tex



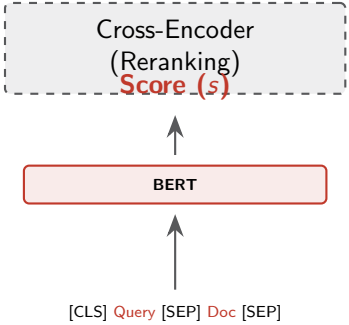
contriever_training.tex

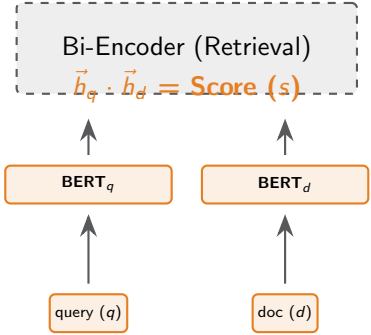


cross_encoder_architecture.tex

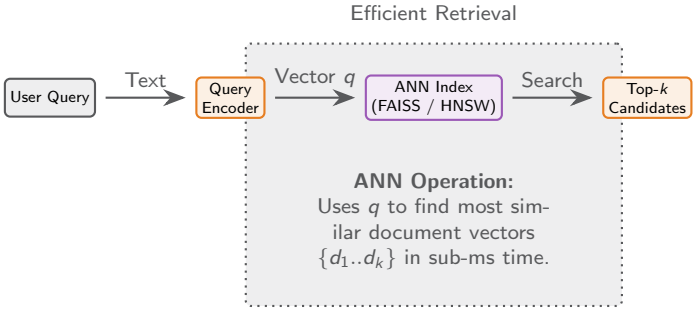


cross_vs_bi_encoder.tex





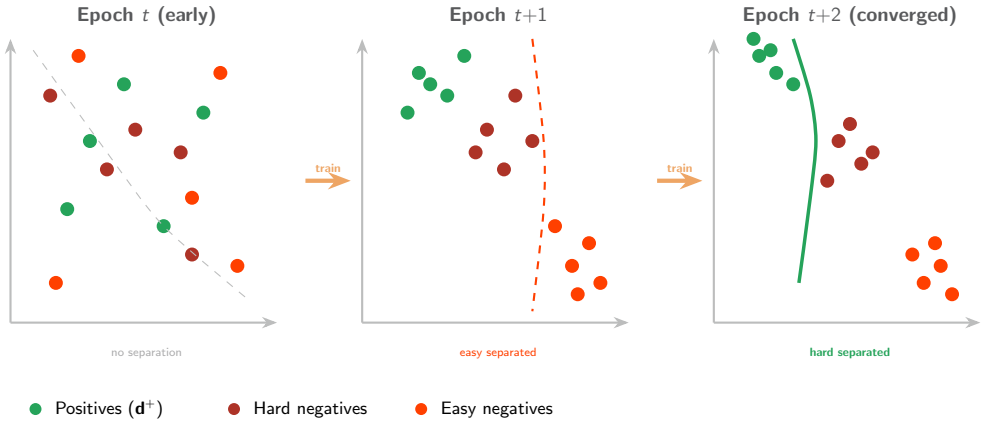
dense_retrieval_pipeline.tex



embedding_matrix.tex

	dim 1	dim 2		dim d	
cat	0.12	-0.5	$\cdots$	0.88	$\vec{w}_{\text{cat}}$
dog	0.15	-0.4	$\cdots$	0.82	
bank	$\vdots$	$\vdots$	$\ddots$	$\vdots$	
word V	0.05	-0.1	$\cdots$	0.44	

evolving_encoder.tex



feature_vector.tex

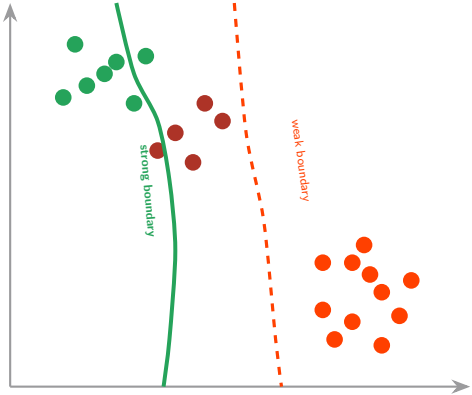
$$\Phi(q, d) = \underbrace{\begin{array}{c|c|c|c|c} x_1 & x_2 & x_3 & \cdots & x_n \\ \text{BM25 Score} & \text{PageRank} & \text{Click Count} & & \text{Freshness} \end{array}}_{\mathbf{x}_{q,d} \in \mathbb{R}^{1000+}}$$

The LTR Goal

Learn $f(\mathbf{x}) \rightarrow$ score

for optimal ranking

hard_negatives_viz.tex



Positives ($\mathbf{d}^+$)
Hard negatives
Easy negatives

InfoNCE Loss:

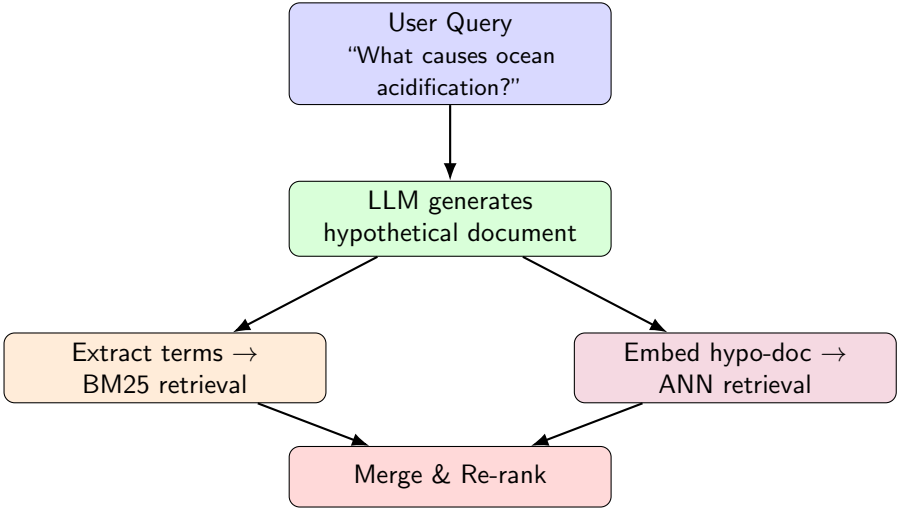
$$\mathcal{L}(\mathbf{q}_i, \mathbf{p}_i^+, \mathbf{p}_{i,1}^-, \cdots, \mathbf{p}_{i,n}^-)$$

$$= -\log \frac{e^{\text{sim}(\mathbf{q}_i, \mathbf{p}_i^+)}}{e^{\text{sim}(\mathbf{q}_i, \mathbf{p}_i^+)} + \sum_{j=1}^{n} e^{\text{sim}(\mathbf{q}_i, \mathbf{p}_{i,j}^-)}}$$

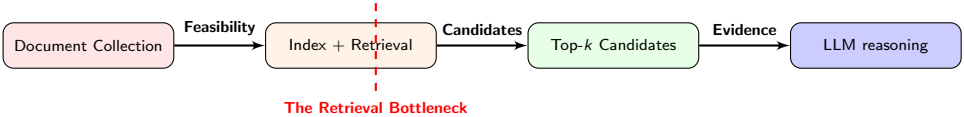
Hard negatives are close to the positives in embedding space $\rightarrow$ large $\text{sim}(\mathbf{q}, \mathbf{p}^-)$.

This makes the denominator large, the loss **harder to minimize**, and gradients **more informative**.

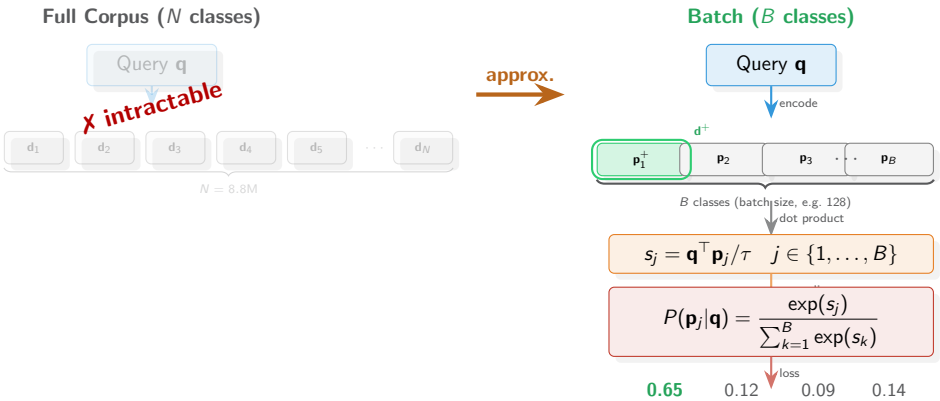
hyde_pipeline.tex



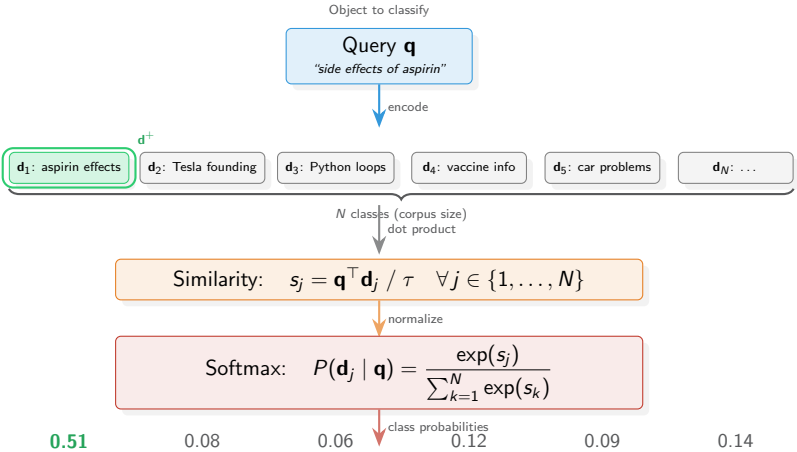
indexing_controls.tex



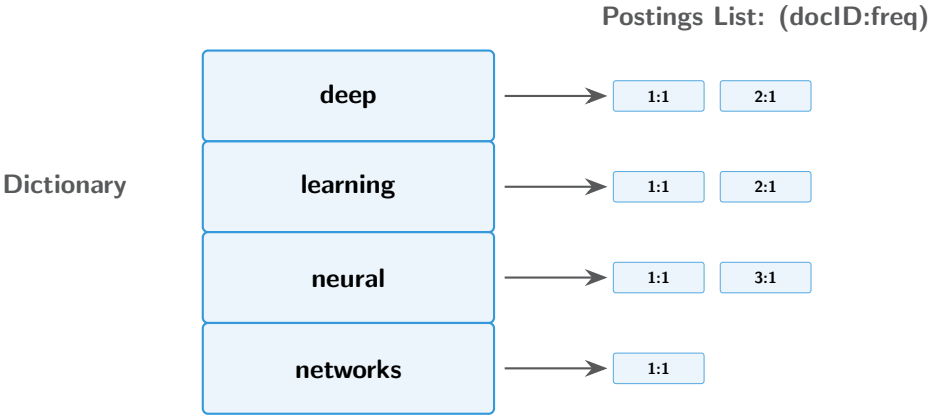
infonce_batch_approx.tex



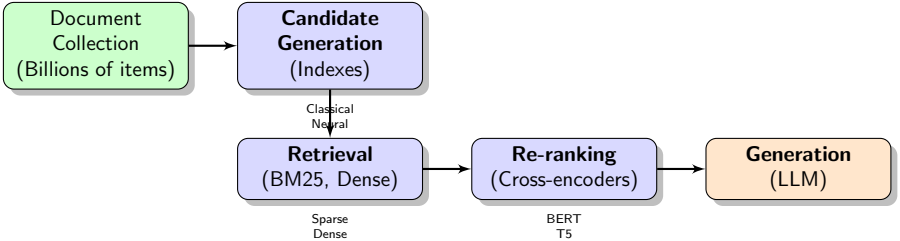
infonce_classification.tex



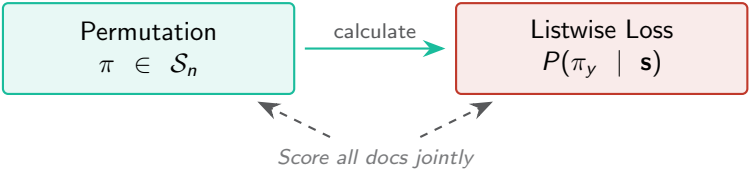
inverted_index_structure.tex



landscape.tex

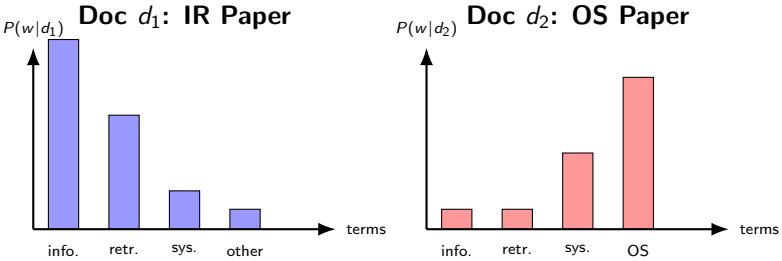


listwise_visualization.tex



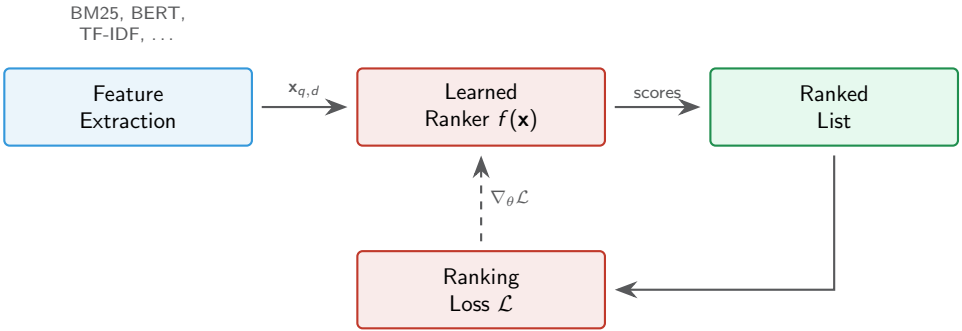
Goal: Minimize distance between predicted and ideal permutations

lm_distribution_viz.tex

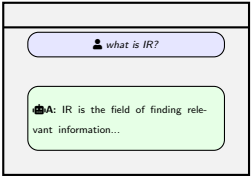


Concept: Each document is a probability distribution over the entire vocabulary.
Retrieval asks: *"How likely is the query sequence under this distribution?"*

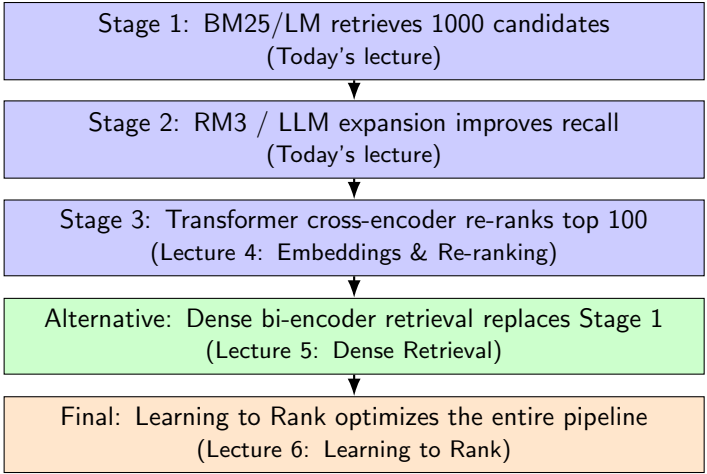
ltr_pipeline.tex



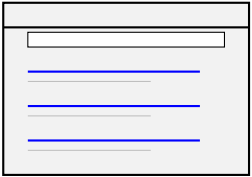
modern_interface.tex



modern_retrieval_stack.tex

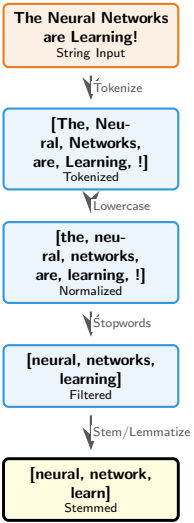


old_interface.tex

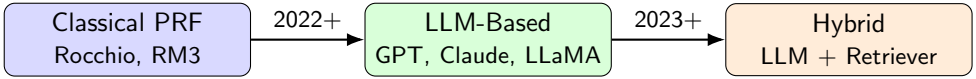


pairwise_concept.tex

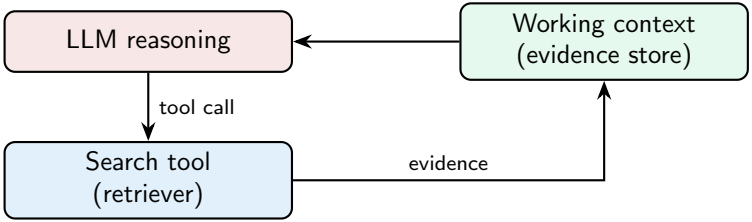
preprocessing_pipeline.tex



query_expansion_timeline.tex



rag_dynamic_search.tex



budgeted: max calls, max tokens

rag_failure_taxonomy.tex

1) **Retrieval**: top-k misses $e^\star$

2) **Selection**: dedup/truncation removes $e^\star$

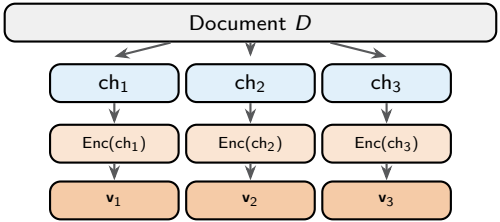
3) **Reasoning**: wrong inference from $e^\star$

4–6) **Grounding**: unsupported claims/citations

Instrument logs at every boundary.
No logs = no debugging.

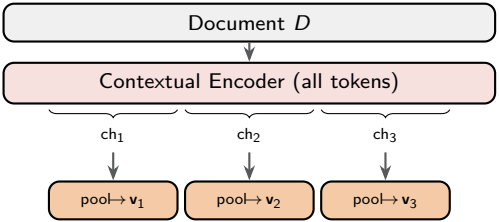
rag_late_chunking.tex

Standard: Chunk → Encode



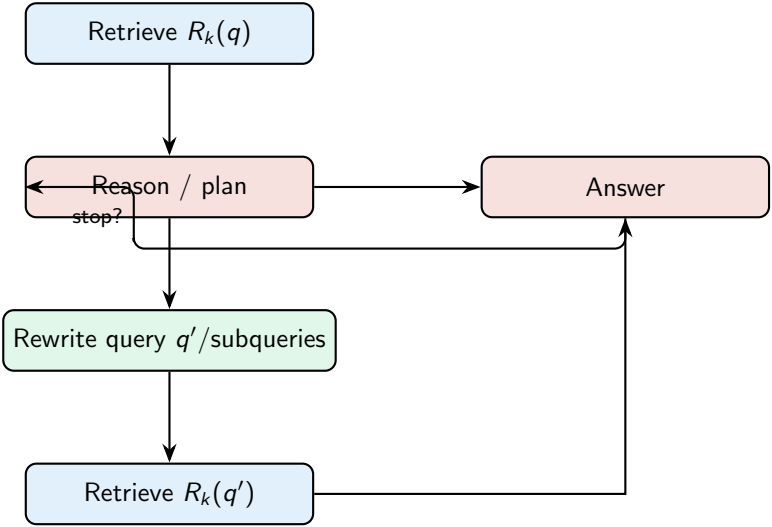
× *isolated: "Paris" in ch_2 unaware ch_1 says "capital of France"*

Late Chunking: Encode → Pool

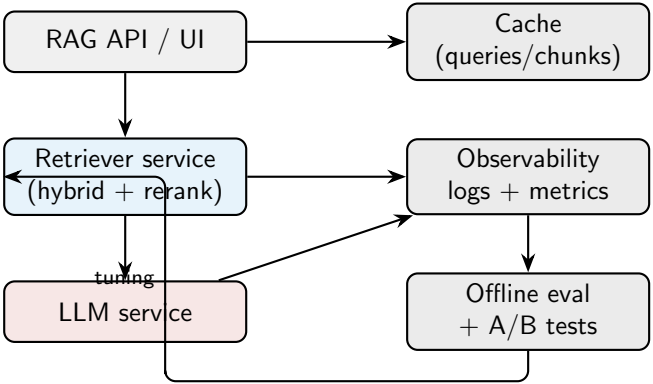


✓ *full document context; same index granularity as standard chunking*

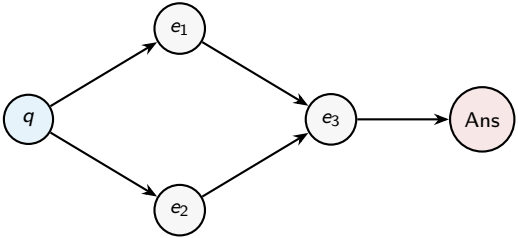
rag_multistep.tex



rag_production_arch.tex

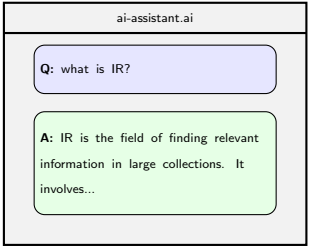


rag_reasoning_chain.tex

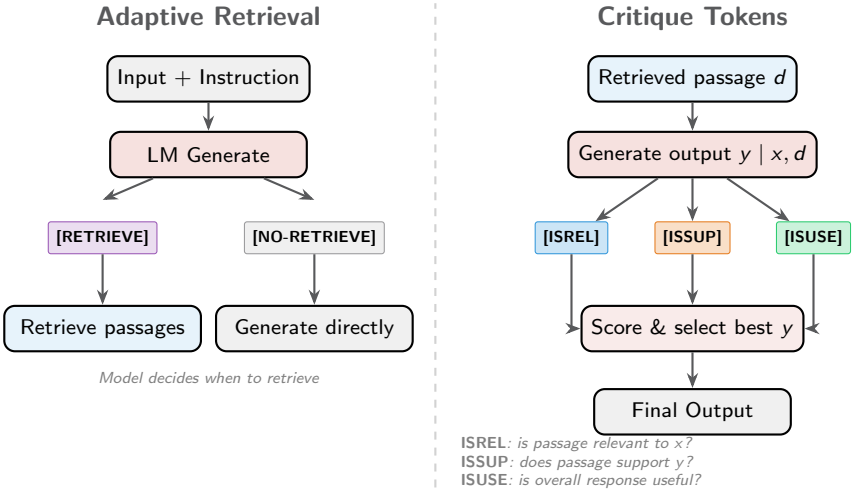


multi-hop: bridge intermediate evidence

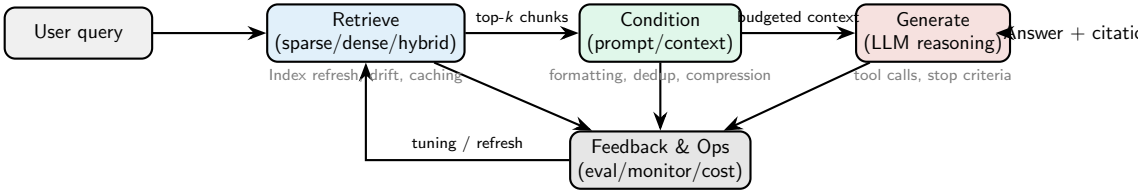
rag_search.tex



rag_selfrag.tex



rag_system_view.tex



representation_gap.tex

Sparse Representation
$[0, 0, \ldots, 3, \ldots, 0, 2, \ldots, 0]$
$\in \mathbb{R}^{|V|} \quad (|V| \approx 30{,}000)$
$>$99.9% zeros

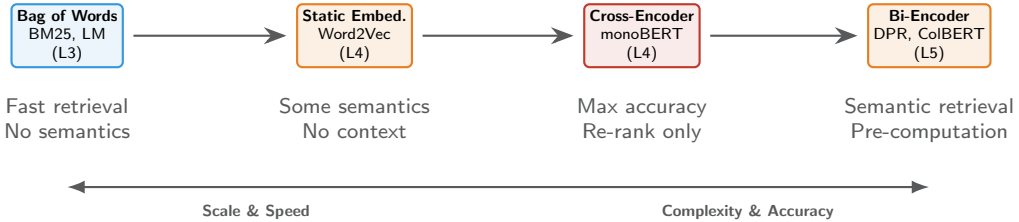
This lecture →

Dense Representation
$[0.21, -0.05, 0.83, \ldots]$
$\in \mathbb{R}^{d} \quad (d = 768)$
100% non-zero

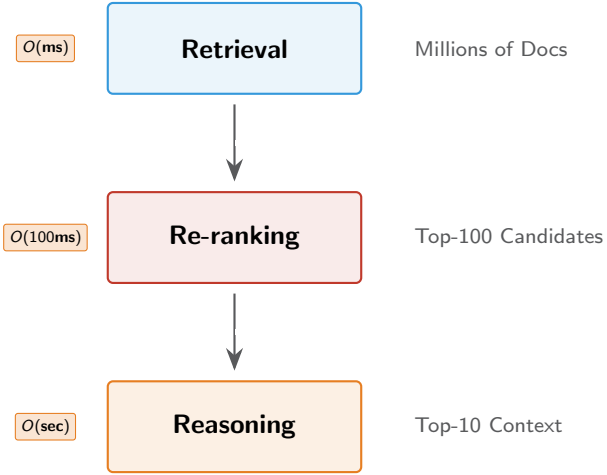
Each dimension = one word
Lexical matching only

Each dimension = latent feature
Semantic matching

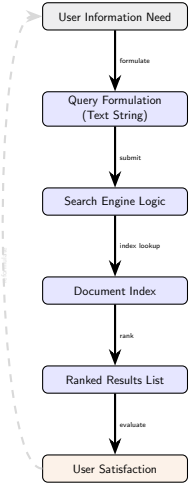
representation_spectrum.tex



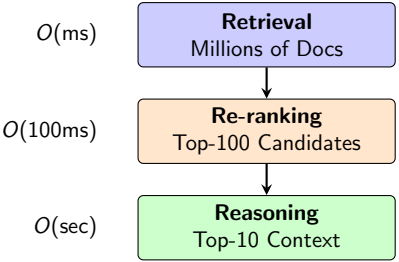
retrieval_reasoning.tex



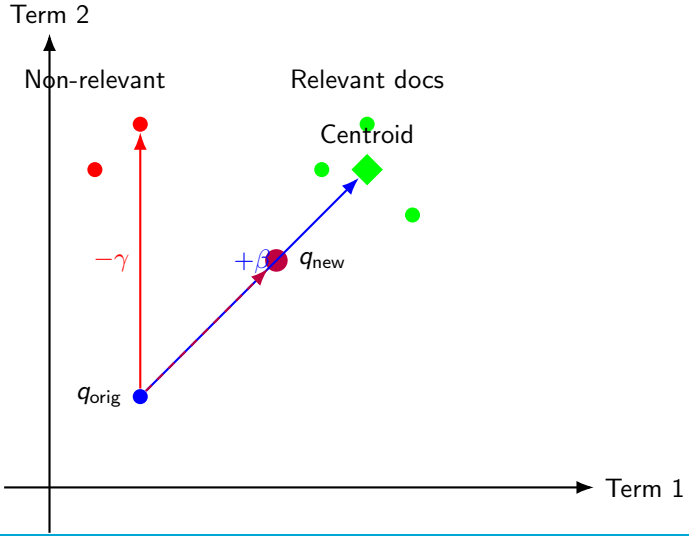
retrieval_reasoning_flow.tex



retrieval_reasoning_standalone.tex



rocchio_geometric.tex



rrf_visualization.tex

BM25 Ranking

① Doc A
② Doc B
③ Doc C

rank-based (no scores)

Dense Ranking

Doc C ①
Doc A ②
Doc D ③

rank-based (no scores)

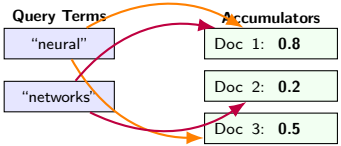
Final RRF Rank

Doc A $\frac{1}{k+1} + \frac{1}{k+2}$
Doc C $\frac{1}{k+3} + \frac{1}{k+1}$
Doc B $\frac{1}{k+2}$

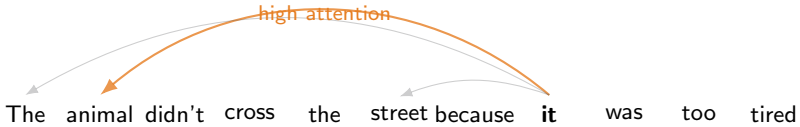
$$Score(d) = \sum_{r \in R} \frac{1}{k + rank(d, r)}$$

typically $k = 60$

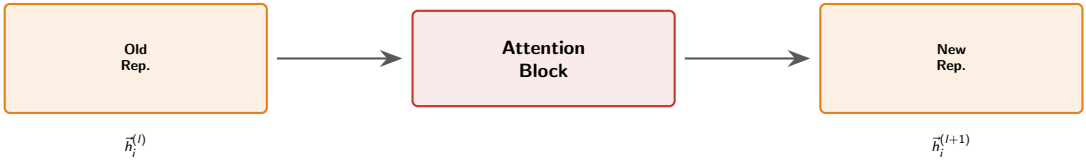
score_aggregation.tex



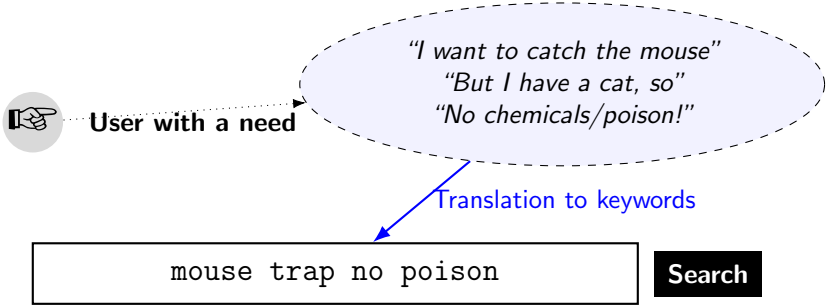
self_attention_example.tex



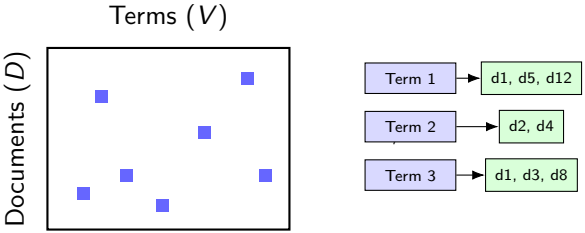
self_attention_logic.tex



semantic_gap_mouse.tex



sparse_matrix.tex



sparse_vs_dense_representation.tex

Sparse Representation

$[0, 0, \ldots, 3, \ldots, 0, 2, \ldots, 0]$

$\in \mathbb{R}^{|V|} \quad (|V| \approx 30k)$

$>$99.9% zeros

This lecture $\longrightarrow$

Dense Representation

$[0.21, -0.05, 0.83, \ldots]$

$\in \mathbb{R}^{d} \quad (d = 768)$

100% non-zero

Each dimension = one word
Lexical matching only

Each dimension = latent feature
Semantic matching

static_vs_contextualized.tex

Static Embeddings

Word2Vec, GloVe

"bank" $\rightarrow$ **one vector** always

Context-independent

2018 Evolution $\longrightarrow$

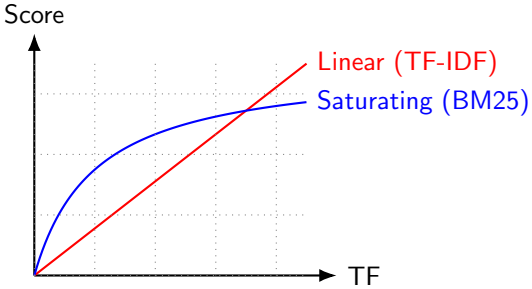
Contextualized Embeddings

BERT, RoBERTa, GPT

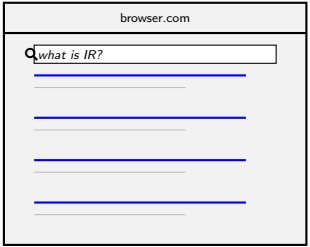
"bank" $\rightarrow$ **dynamic vector** depending on neighbors

Context-dependent

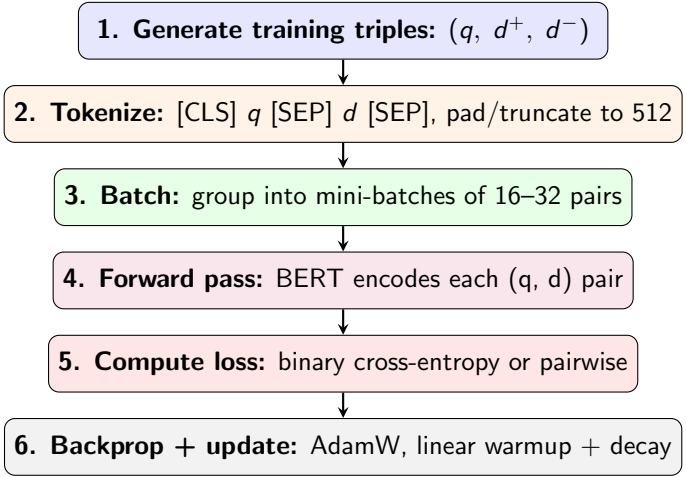
tf_saturation_comparison.tex



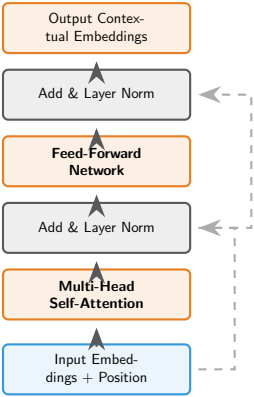
traditional_search.tex



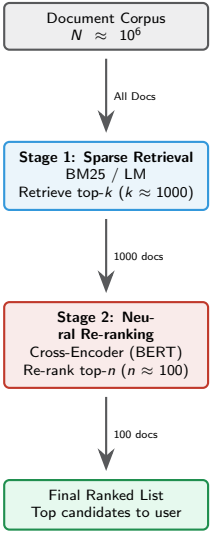
training_pipeline.tex



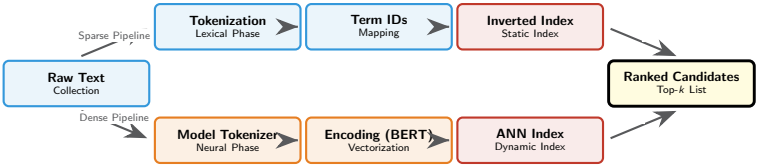
transformer_encoder_block.tex



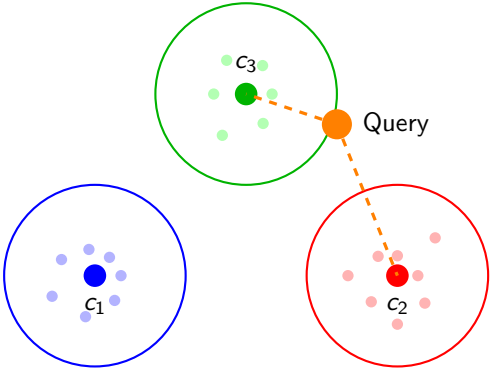
two_stage_pipeline.tex



unified_view.tex

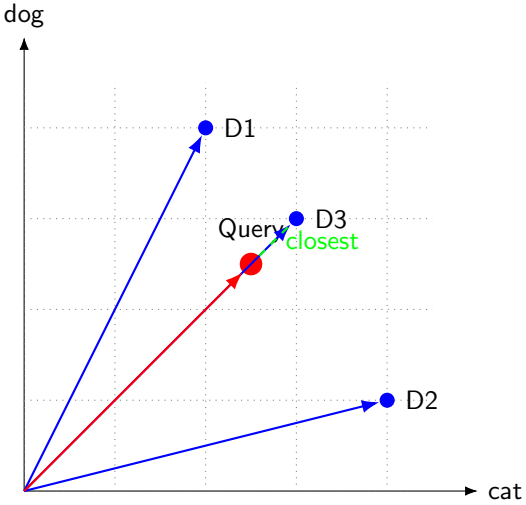


vector_projection.tex

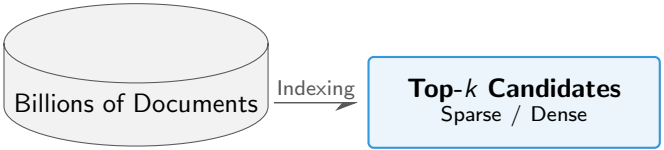


Search only in 1-2 nearest clusters (not all!)

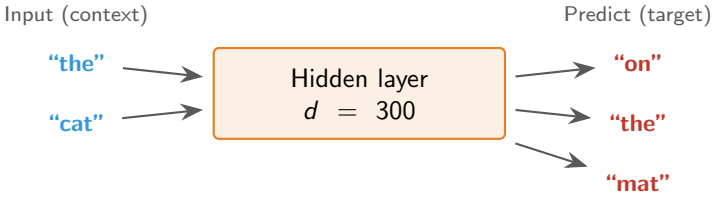
vsm_geometric_view.tex



why_indexing.tex



word2vec_skipgram.tex



After training: Weights reveal semantic word vectors.
Words in similar contexts $\rightarrow$ similar vectors.

wordpiece_example.tex

WordPiece Example (BERT):

[‘‘un’’, ‘‘##touch’’, ‘‘##able’’]

Breaks word into sub-tokens based on statistical frequency.